

Biochemistry Lec.4

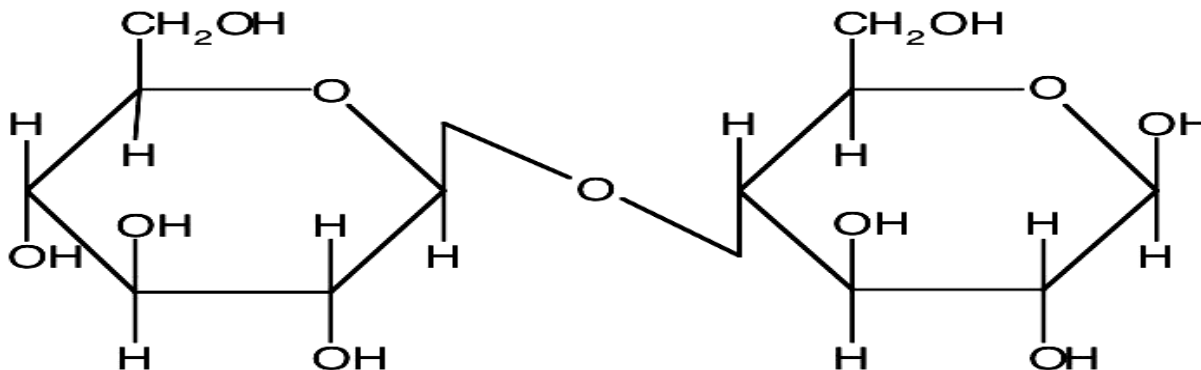
- **Disaccharides**
- **Homopolysaccharides**



Disaccharides

5- Cellobiose

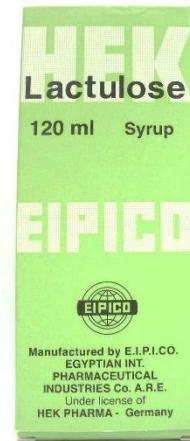
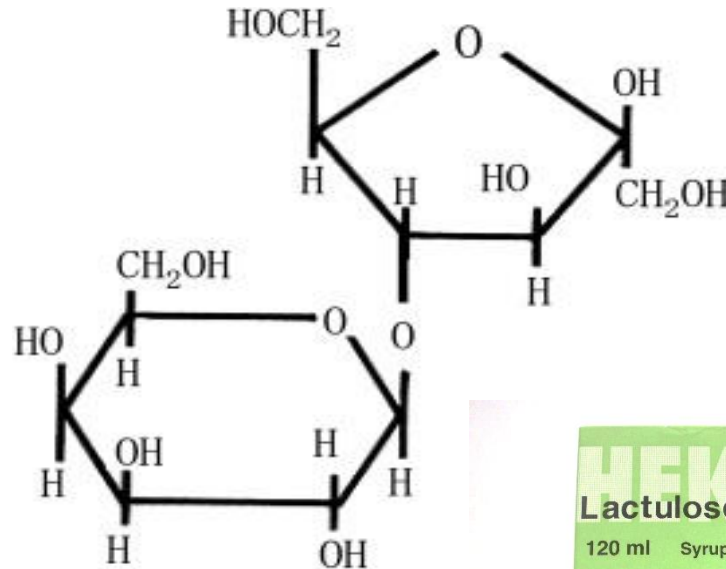
- ✓ It is formed of 2 units of β -D-glucopyranose linked by β 1-4 glycosidic bond.
- ✓ It is obtained by partial hydrolysis of cellulose by cellulase enzyme in animals.
- ✓ It is not digested in human as they are deficient in cellulase enzyme.
- ✓ It is a reducing sugar.
- ✓ It can be present in α and β forms.
- ✓ It can show mutarotation.



Disaccharides

6- Lactulose

- ✓ It is formed of β -D-galactopyranose and β -D-fructofuranose linked by β 1-4 glycosidic bond.
- ✓ It is present in plant.
- ✓ It is reducing sugar.
- ✓ It can be present in α and β forms.
- ✓ It can show mutarotation.
- ✓ It is prepared by alkaline rearrangement of lactose
- ✓ It has therapeutic importance:
 - Osmolar laxative relief chronic constipation.
 - Relief hyperammonemia in hepatic encephalopathy.



Polysaccharides

1- Homopolysaccharides

1. Storage polysaccharides

- **Glucosan:** starch, glycogen, dextrin and dextran.
- **Fructosan:** inulin

2. Structural polysaccharides

- **Cellulose:** contain only β -glucose units.
- **Chitin:** contains β -N-acetyl glucosamine.

2- Heteropolysaccharides

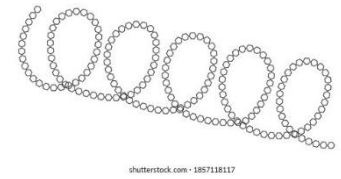
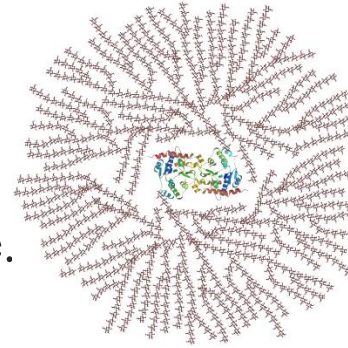
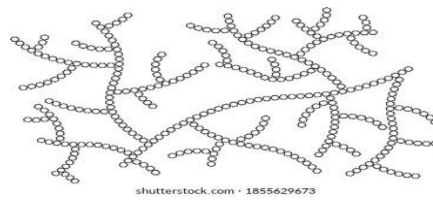
1. Acidic (mucopolysaccharides = glycosaminoglycans = GAGs)

- **Sulfate free** e.g. hyaluronic acid.
- **Sulfated** e.g. chondroitin sulfate. Heparin, dermatan sulfate, keratin sulfate and heparan sulfate

2. Neutral polysaccharides

ex: Blood group – glycoprotein hormones

Starch



Starch is composed of repeating unit of glucose.

It is a mixture of amylose and amylopectin and is found in plant foods.

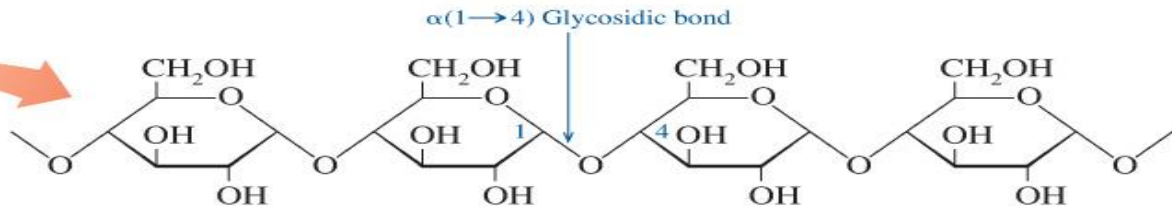
Amylose and amylopectin

Amylose makes 15 - 20% of plant starch and is made up of 250–4000 α D-glucose units bonded $\alpha(1\rightarrow4)$ in a continuous chain (unbranched).

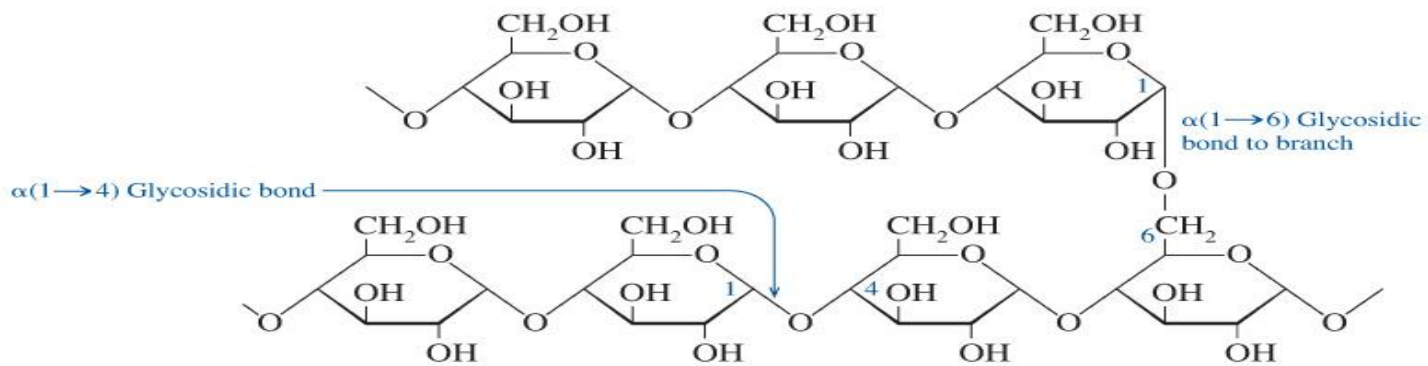
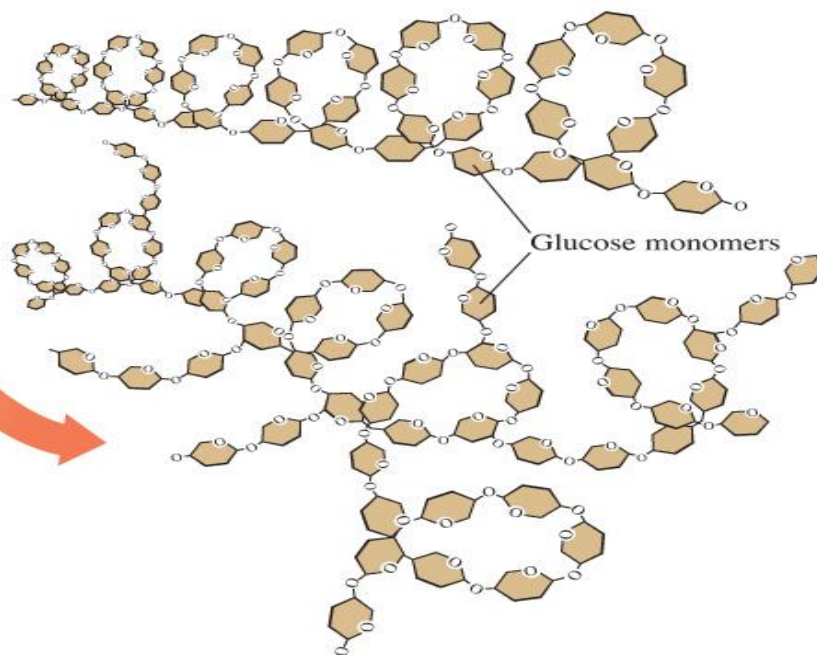
Long chains of amylose tend to coil.

Amylopectin makes 80 - 85% of plant starch and is made up of 7000 α D-glucose units connected by $\alpha(1\rightarrow4)$ glycosidic bonds.

Amylopectin is branched chain: the glucose unit is connected at the branching point by $\alpha(1\rightarrow6)$ glycosidic bonds at every 25-30 glucose units.



a. Unbranched chain of amylose



b. Branched chain of amylopectin

- ❖ Starch is white, odorless, tasteless powder .
- ❖ Starch is insoluble in cold water, but solubilized on heating and form gel /paste.
- ❖ Starch is a non reducing carbohydrate (Negative Benedicts Test).
- ❖ Starch on Iodine reaction gives blue color.
- ❖ Starch is a storage form of glucose and serves as reservoir of energy in plants.
- ❖ In humans, it is a predominant form of dietary carbohydrate ingested through foods which has high dietary and calorific value.
- ❖ digested in mouth by salivary amylase and in intestine by pancreatic amylase.
- ❖ α Amylase cleaves, α (1-4) glycosidic bonds of Amylose and Amylopectin and releases Maltose and Isomaltose.
- ❖ Maltose and Isomaltose is then digested by Maltase and Isomaltase to release free Glucose units.
- ❖ Thus Starch on digestion gives thousands of free Glucose units which have high calorific value.

Glycogen = Animal starch

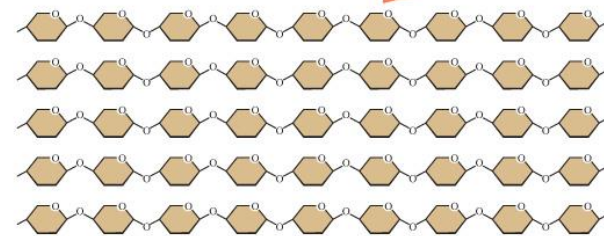
- ✓ **Glycogen** is a storage polysaccharide found in animals.
- ✓ Glycogen is stored in the liver (75 gm) and muscles (125 gm).
- ✓ It is composed of 25,000-30,000 glucose units linked by α (1 $\rightarrow$ 4) and α (1 $\rightarrow$ 6) glycosidic bonds.
- ✓ Its structure is similar to amylopectin, except that α (1 $\rightarrow$ 6) branching occurs about every 12 glucose units (highly branched).
- ✓ Glycogen is sparingly soluble in water.
- ✓ On Iodine reaction glycogen gives deep red color.
- ✓ Biomedical importance:
 1. Glycogen is a storage form of glucose in animal and human body after well fed conditions.
 2. It serves as reservoir of glucose which can be used in emergency conditions (Fasting /Between meals)
- ✓ Dietary glycogen in GIT is digested by α - Amylase to Maltose and Isomaltose and finally to thousands of Glucose Units.

Cellulose

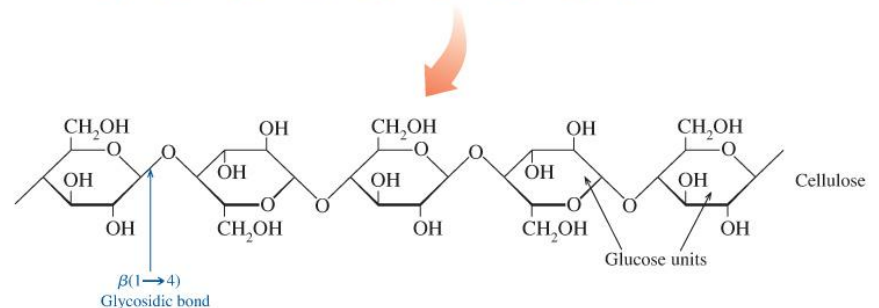
- 1) **Cellulose** contains glucose units (2500-14000) bonded β (1 $\rightarrow$ 4).
- 2) The chain of glucose units is straight. This allows chains to align next to each other to form a strong rigid structure.
- 3) In plants cellulose present in cell wall provides structural and mechanical support.
- 4) In human, it is non digestible carbohydrate serve as dietary fiber.



Fibers formed from aligned cellulose



Cellulose aligns to form rigid structures



Biomedical importance of cellulose in human:

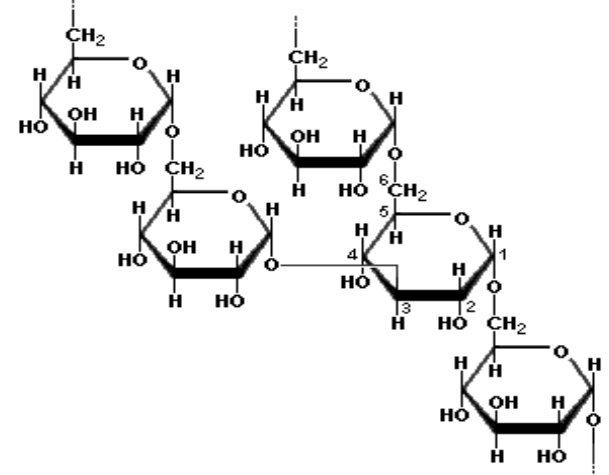
- 1- Eliminates daily, metabolic wastes and toxins out of the body, through feces.
- 2- Cellulose prevents constipation, and reduces risk of :
 - colon cancer
 - varicose veins
 - hemorrhoids
- 3- Cellulose reduces the absorption of :
 - Glucose prevent Diabetes mellitus
 - Cholesterol prevent Atherosclerosis
- 4- Cellulose being non caloric and possessing high satiety value helps in managing obesity in humans (adult 20-25gm/day).
- 5- Its disadvantage decreases absorption of minerals.

Dextrin

- It is an intermediate hydrolytic product of Starch digestion
- Its repeating Unit - α D Glucose.
- Dextrin is less complex than starch structure.
- Types of dextrin:
 1. Amylodextrin gives violet to Iodine reaction
 2. Erythrodextrin gives red color to Iodine reaction
 3. Achrodextrin gives colorless to Iodine reaction.
- In human GIT Dextrin is obtained as an intermediate hydrolytic product of Starch digestion by the action of α -Amylase activity.
- Biomedical importance of dextrin:
 1. Dextrin has sticky nature hence used as binding and adhesive agent.
 2. Dextrin is used as infant food.

Dextran

- Dextran is a Bacterial Homopolysaccharide
- Its repeating Unit – α D Glucose.
- It is highly branched, complex, of glucose units linked by α (1-6) and α (1-3) glycosidic linkages.



- Dextran is obtained from **Sucrose** solution incubated with bacteria.
- **Biomedical importance of dextran:**
 1. Dextran solution is **high molecular weight**, **viscous** solution with **osmotic pressure** equivalent to **plasma Albumin**.
 2. Dextran **maintains blood volume** and **osmotic pressure**.
 3. **Dextran solution** is used as **Plasma substitute/plasma volume expander**.
 4. Dextran infusion manages a **hemorrhagic case** and **prevent** from **hypovolemic shock**.

Inulin

- Inulin is a Fructosan.
- Its repeating unit is β , D Fructose units (33-35 units) linked by β (2-2) glycosidic bonds.
- Inulin is non digestible and non absorbable form in human GIT, hence have no calorific value.
- **Biomedical importance of inulin:**
 1. Inulin solution infused intravenously during Inulin Clearance Test, checks Glomerular Filtration Rate (GFR) of Kidney (120-125 ml/min).
 2. Improvement of immune response.
 3. Reduce colon cancer risk.
 4. It has antioxidant activity.

Chitin

- Chitin makes up the exoskeleton of insects and crustaceans and cell walls of some fungi.
- It is made up of N-acetylglucosamine containing β (1 $\rightarrow$ 4) glycosidic bonds.
- It is structurally strong.
- **Biomedical importance of chitin:**
 1. Chitin is used as surgical thread that biodegrades as a wound heals.
 2. It serves as a protection from water in insects.
 3. Chitin is used in waterproof paper, cosmetics and lotions to retain moisture.

